Quantum Field Theory

Topic 1: Quantization of Klein-Gordon Field

Ladder Operators and the Number Operator

In harmonic oscillators, a and $a^\dagger$ are lowering and raising operators. a reduces $|n\rangle$ to $|n-1\rangle$, annihilating a quantum, while $a^\dagger$ raises $|n\rangle$ to $|n+1\rangle$, creating a quantum.
We define the **Number Operator** N and its eigenstates $|n\rangle$:

$$N = a^\dagger a, \quad N|n\rangle = n|n\rangle$$

The operators satisfy the fundamental commutation relation:

$$[a, a^\dagger] = 1 \implies \mathbf{aa^\dagger = N + 1}$$

Action of the Raising Operator

To find the effect of $a^\dagger$ on $|n\rangle$, we apply the number operator N:

$$\begin{aligned} N(a^\dagger|n\rangle) &= (a^\dagger a)a^\dagger|n\rangle = a^\dagger(aa^\dagger)|n\rangle \\ &= a^\dagger(N+1)|n\rangle \quad \text{(Using } aa^\dagger = N+1\text{)} \\ &= a^\dagger(N|n\rangle + |n\rangle) = a^\dagger(n|n\rangle + |n\rangle) \\ &= \mathbf{(n+1)a^\dagger|n\rangle} \end{aligned}$$

Conclusion

Applying $a^\dagger$ to $|n\rangle$ results in an eigenstate of N with eigenvalue $(n+1)$.

Action of the Lowering Operator

To find the effect of a on $|n\rangle$, we apply the number operator N to the state $(a|n\rangle)$:

$$\begin{aligned} N(a|n\rangle) &= (a^\dagger a)a|n\rangle \\ &= (aa^\dagger - 1)a|n\rangle \quad \text{(Using } [a, a^\dagger] = 1) \\ &= a(a^\dagger a)|n\rangle - a|n\rangle \\ &= a(N|n\rangle) - a|n\rangle \\ &= a(n|n\rangle) - a|n\rangle = \mathbf{(n-1)a|n\rangle} \end{aligned}$$

Conclusion

Applying a to $|n\rangle$ results in a new eigenstate of N with eigenvalue $(n-1)$.

Expressions for a and $a^\dagger$

Using the adjoint property to find normalization constants:

- Raising Operator: $\|a^\dagger|n\rangle\|^2 = \langle n|aa^\dagger|n\rangle = n+1$
- Lowering Operator: $\|a|n\rangle\|^2 = \langle n|a^\dagger a|n\rangle = n$

Final Normalized Results

$$\mathbf{a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle} \quad \text{and} \quad \mathbf{a|n\rangle = \sqrt{n}|n-1\rangle}$$

Extension to Field Quantization

In field quantization, operators are indexed by wave vector k:

- **Quantization in a box:** $[a_k, a_{k'}^\dagger] = \delta_{k,k'}$
- **Continuous limit:** $[a_k, a_{k'}^\dagger] = (2\pi)^3 \delta^3(k - k')$

Harmonic Oscillator Hamiltonian

Starting with the classical Hamiltonian $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$:

$$a = \frac{1}{\sqrt{\hbar\omega}}\left(\frac{ip}{\sqrt{2m}} + \sqrt{\frac{m}{2}}\omega q\right), \quad a^\dagger = \frac{1}{\sqrt{\hbar\omega}}\left(\frac{-ip}{\sqrt{2m}} + \sqrt{\frac{m}{2}}\omega q\right)$$

Calculating the product $\hbar\omega a^\dagger a$ yields:

$$\hbar\omega a^\dagger a = \frac{p^2}{2m} + \frac{m}{2}\omega^2 q^2 + \frac{i\omega}{2}[p,q] = H + \frac{i\omega}{2}[p,q]$$

Result: Quantum Harmonic Oscillator

$$\mathbf{H} = \left(\mathbf{a}^\dagger \mathbf{a} + \tfrac{1}{2}\right)\hbar\omega$$

Klein-Gordon Lagrangian: Fourier Expansion

We begin with the field Lagrangian density and expand the field $\phi(x)$ into Fourier modes in a box of volume V:

$$L = \int d^3x \left[\frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2 \right]$$

Substituting the expansion $\phi(x,t) = \frac{1}{\sqrt{V}} \sum_k q_k(t) e^{i\vec{k}\cdot\vec{x}}$ yields:

$$L = \frac{1}{2}\dot{q}_0^2 - \frac{1}{2}\omega_0^2 q_0^2 + \sum_{k>0} (\dot{q}_k \dot{q}_{-k} - \omega_k^2 q_k q_{-k})$$

where the dispersion relation is $\omega_k^2 = k^2 + m^2$.

Klein-Gordon Hamiltonian

We define the conjugate momentum $p_k = \frac{\partial L}{\partial \dot{q}_k} = \dot{q}_{-k}$, implying $p_0 = \dot{q}_0$. Using the Legendre transformation $H = \sum p_k \dot{q}_k - L$:

$$H = \frac{1}{2}p_0^2 + \frac{1}{2}\omega_0^2 q_0^2 + \sum_{k>0}(p_k p_{-k} + \omega_k^2 q_k q_{-k})$$

By substituting the creation/annihilation operators $a_k = \frac{1}{\sqrt{2\hbar\omega_k}}(\omega_k q_k + i p_{-k})$, we reach the quantized form:

$$H = \sum_k \hbar\omega_k \left(a_k^\dagger a_k + \frac{1}{2}\right) \xrightarrow{\text{relative}} \sum_k \hbar\omega_k a_k^\dagger a_k$$

Comparing Discrete QM vs. Field Theory

Feature	**Harmonic Oscillator**	**Klein-Gordon Field**
State $\|n\rangle$	n-th energy level	State with n particles
Lowest State	Ground state	Vacuum state $\|0\rangle$
Excitation	Increasing energy	Creating a particle
Energy	$(n+\frac{1}{2})\hbar\omega$	$\sum_k \hbar\omega_k a_k^\dagger a_k$

In the field case, we redefine zero-point energy so that $H|0\rangle = 0$.

Multi-particle States and Relativistic Relation

Multi-particle states are denoted by $|\Psi\rangle = a^\dagger_{k_n} \dots a^\dagger_{k_1}|0\rangle$.

- **Total Momentum:** $p|\Psi\rangle = \hbar(\vec{k}_1 + \cdots + \vec{k}_n)|\Psi\rangle$
- **Total Energy:** $H|\Psi\rangle = \hbar(\omega_{k_1} + \cdots + \omega_{k_n})|\Psi\rangle$

The frequency ω_k follows the relativistic relation:

$$\hbar\omega_{\mathbf{k}} = \sqrt{\hbar^2\mathbf{k}^2 + \mathbf{m}^2}$$

General Field Quantization

To quantize a field $\phi(t, \vec{x})$, we expand it into plane wave modes:

$$\phi(t, \vec{x}) = \mathcal{N} \sum_k q_k(t) e^{i\vec{k}\cdot\vec{x}}$$

- $\mathcal{N}$: Normalization factor.
- $q_k(t)$: Harmonic oscillator coordinates in mode space.
- $\vec{k}$: Wave vector, which corresponds to particle momentum after quantization.

Under periodic boundary conditions in a box of length L, the wave vector is discrete: $\vec{k} = \frac{2\pi}{L}(m_1, m_2, m_3)$.

Quantization of the Klein-Gordon Field

We begin with the general field expansion into plane waves under periodic boundary conditions:

$$\phi(t, \vec{x}) = \mathcal{N} \sum_k q_k(t) e^{i\vec{k}\cdot\vec{x}}$$

Substituting this into the Klein-Gordon wave equation $(\hbar^2 \partial_\mu \partial^\mu + m^2)\phi = 0$:

$$\hbar^2 \partial_\mu \partial^\mu (\mathcal{N} \sum_k q_k(t) e^{i\vec{k}\cdot\vec{x}}) + m^2 (\mathcal{N} \sum_k q_k(t) e^{i\vec{k}\cdot\vec{x}}) = 0$$

$$\hbar^2 \mathcal{N} \left(\sum_k \ddot{q}_k(t) e^{i\vec{k}\cdot\vec{x}} - \sum_k q_k(t) k^2 e^{i\vec{k}\cdot\vec{x}} \right) + m^2 \mathcal{N} \sum_k q_k(t) e^{i\vec{k}\cdot\vec{x}} = 0$$

$$\left(\hbar^2 \sum_k \ddot{q}_k(t) + \sum_k k^2 q_k(t) + m^2 \sum_k q_k(t) \right) e^{i\vec{k}\cdot\vec{x}} = 0$$

Since $e^{i\vec{k}\cdot\vec{x}}$ are linearly independent, each mode must satisfy:

$$\boxed{\ddot{q}_k(t) + \omega_k^2 q_k(t) = 0, \quad \text{where } \omega_k = \sqrt{k^2 + \frac{m^2}{\hbar^2}}}$$

Real Klein-Gordon Field Solution

The most general solution for the mode harmonic oscillator is $q_k(t) = A_k e^{-i\omega_k t} + B_k e^{i\omega_k t}$. For a real field, $\phi(t, \vec{x}) = \phi^*(t, \vec{x})$, implying $q_k(t) = q^*_{-k}(t)$:

$$\begin{aligned} q^*_{-k}(t) &= A^*_{-k} e^{i\omega_k t} + B^*_{-k} e^{-i\omega_k t} \\ \implies B_k &= A^*_{-k} \end{aligned}$$

Substituting B_k back and converting constants to operators ($A_k \to a_k$ and $A^*_{-k} \to a^\dagger_{-k}$):

$$q_k(t) = a_k e^{-i\omega_k t} + a^\dagger_{-k} e^{i\omega_k t}$$

After normalization, the mode coordinate becomes:

$$\boxed{q_k(t) = \frac{1}{\sqrt{2\hbar\omega_k}} \left(a_k e^{-i\omega_k t} + a^\dagger_{-k} e^{i\omega_k t} \right)}$$

Removing the Box: Continuous Limit

As the box volume $V \to \infty$, the discrete sum becomes an integral $\int \frac{d^3k}{(2\pi)^3} q_k(t) e^{i\vec{k}\cdot\vec{x}}$:

$$\phi(t,\vec{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_k e^{-i\omega_k t} e^{i\vec{k}\cdot\vec{x}} + a^\dagger_{-k} e^{i\omega_k t} e^{i\vec{k}\cdot\vec{x}} \right)$$

Applying a dummy variable $-k$ to the second term:

$$\boxed{\phi(t,\vec{x}) = \int \frac{d^3k}{(2\pi)^3\sqrt{2\omega_k}} \left(a_k e^{-ik\cdot x} + a^\dagger_k e^{ik\cdot x} \right)}$$

Physical Interpretation

- a_k: Destruction operator multiplying positive frequency modes.
- $a^\dagger_k$: Creation operator multiplying negative frequency modes.
- $k \cdot x = \omega_k t - \vec{k} \cdot \vec{x}$.

Canonical Commutation and Quantization

To complete the quantization of the field, we define the canonically conjugate field momentum $\Pi(t, \vec{x})$ as the Fourier transform of the mode momentum P_k:

$$\Pi(t, \vec{x}) = \int \frac{d^3k}{(2\pi)^3} P_k e^{i\vec{k}\cdot\vec{x}}$$

where the canonical momentum for each mode is derived from the Lagrangian:

$$\begin{aligned} P_k &= \frac{\partial L}{\partial \dot{q}_k} = \dot{q}_{-k} \\ &= -i\sqrt{\frac{\omega_k}{2}} \left(a_k e^{-i\omega_k t} - a_k^\dagger e^{i\omega_k t} \right) \end{aligned}$$

The field satisfies the Equal-Time Commutation Relation (Quantization Condition):

$$\boxed{[\phi(t, \vec{x}), \Pi(t, \vec{x}')] = i\hbar\delta^3(\vec{x} - \vec{x}')}$$

Conclusion

A field is considered to be quantized when its canonical variables satisfy a commutation relation that is non-zero and specifically proportional to $i\hbar$.

Extension to Complex Fields

Complex fields describe particles carrying charge, requiring two sets of oscillators ($a, a^\dagger$ and $b, b^\dagger$) to represent particles and anti-particles:

$$\boxed{\phi(t,\vec{x}) = \int \frac{d^3k}{(2\pi)^3\sqrt{2\omega_k}} \left(b_k e^{-ik\cdot x} + a_k^\dagger e^{ik\cdot x}\right)}$$

$$\boxed{\phi^\dagger(t,\vec{x}) = \int \frac{d^3k}{(2\pi)^3\sqrt{2\omega_k}} \left(a_k e^{-ik\cdot x} + b_k^\dagger e^{ik\cdot x}\right)}$$

Physical Interpretation of Charges

- In ϕ: b_k destroys a particle of charge -1 and $a_k^\dagger$ creates a particle of charge $+1$. Both produce a net charge of $+1$.
- In $\phi^\dagger$: a_k destroys a particle of charge $+1$ and $b_k^\dagger$ creates a particle of charge -1. Both produce a net charge of -1.
- $b, b^\dagger$ describe anti-particles relative to $a, a^\dagger$ which describe particles.